

SINKGRAB · REQUIREMENTS

SinkGrab requirements

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DOCUMENT CONTROL

Document ID	SKG-REQ-001	Version	0.4
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sinkgrab		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-03	Amish Chadha	Status after Amish's round 2 decisions (SKG-DDR-003, 1A, 2B, 3B) from SKG-CAL-001 v0.2; targets unchanged; R9 now not met by 0.1 kg
0.3	2026-10-03	Amish Chadha	TRL 3 status from SKG-CAL-001 on the constructable design (SKG-DDR-002); R1 and R3 at risk and R4 not met, each posed to Amish as a decision in SKG-DEC-001
0.2	2026-10-03	Amish Chadha	TRL 2 requirements with concept status (SKG-DDR-001); R11 restated against the value-engineering target
0.1	2026-09-30	Amish Chadha	Initial scaffold

SinkGrab requirements

Six of the eleven requirements are met on paper or by design, one can only be shown in a field trial, two are at risk and two are not met (SKG-CAL-001 v0.2, section M). Amish decided the three round 1 requirement decisions on 2026-10-03 (SKG-DDR-003): a third person now cranks during the hoist (R4), the grab closes through a 3:1 tackle (R3) and the scraper toes cut 40 mm past the ring (R1). The cycle time (R4) is still the main miss, at about 3.16 min, 5 % over the 3 min target. The bite (R3) still depends on how full the grab comes up and the depth gained (R1) on how much the soil grips the sinking rings; both are settled in trials. R9 is now missed by 0.1 kg because the longer scraper arms bring the scraper head to 25.1 kg; that, the scraper's fold travel and the wording of R4 are posed to Amish as new questions in SKG-DEC-001 and docs/REVIEW.md . "Met on paper" means shown by calculation, not by test.

Safety: SinkGrab is lifting equipment worked over an open well shaft. R7 (proof load) and R8 (nobody in the well) are its safety requirements, and the overload limits behind R7 (a crank shear pin and a shear link, SKG-CAL-001, section C) are checked on paper only. Every lifting part must be proof-loaded before use (SKG-BLD-001, section 6).

Table 1. Requirements and status at TRL 3

ID	Requirement	Target	Verification (TRL 3 or later)	Status (SKG-CAL-001)
R1	Deepen a flooded caisson well below the water table without de-watering	At least 3 m of standing water depth reached (target)	Field trial in a partner well, measured with the dip tape before and after	At risk: the 8-ring string sinks only while skin friction stays below about 1.7 kPa; estimated 1 to 3 kPa; the toes now cut 40 mm past the ring's outer face to loosen the soil against it (SKG-DDR-003)
R2	Fit common caisson ring sizes	Works inside rings of 0.8 to 1.3 m internal diameter	Fit check in rings of three sizes	Met by design: grab 468 mm across; scraper folds to 377 mm radius with a 320 mm lift; toes, skids and a longer arm pair cover the range. The stop collar limits the lift to 30 mm as modelled (new question in SKG-DEC-001)
R3	Grab load per cycle	At least 20 L of saturated sand per bite (target)	Test pit trials with a measured tipping bin	At risk: 21.2 L at an assumed 75 % fill; 14.1 L at 50 %; the 3:1 tackle closes the lips with about 687 N, 50 % more (SKG-DDR-003)
R4	Cycle time	3 min or less per grab cycle at 10 m depth with two operators (target)	Timed trial over 20 cycles	Not met on paper: about 3.16 min with a third person at the cranks for the hoist (SKG-DDR-003), 5 % over; 3.8 min with two
R5	Operator effort	Crank force 150 N or less per operator at rated grab load (target)	Spring balance on the crank handle during lifts	Met on paper: 54 N each with two people (108 N for one); 36 N each on average with three
R6	Keep the lining vertical while sinking	Tilt no worse than 1 in 80, the allowance used for sunk well foundations (Construction Civil)	Four-point dip tape readings every cycle during trials	Not verifiable at TRL 3: readings resolve about 1 in 250; correction by scraping the high side and saddles
R7	Proof load of lifting parts	Tripod, capstan, hooks and rigging proof-loaded to 2 times working load (target)	CalRig proof-load test with record	Met on paper: proof 193 kg, inside the HatchSide proof of 225 kg; line factor 10.4; shear link margin 1.15
R8	No person in the well during digging	100% of digging and undercutting done from the surface	Trial log and method statement review	Met by design: every task from the surface; doors cover the shaft
R9	Portability	Heaviest single piece 25 kg or less; kit fits in a small pickup (target)	Weigh each piece; load trial	Not met on paper by 0.1 kg: heaviest piece 25.1 kg (scraper head, with the longer arms); kit about 456 kg, longest piece 2.0 m
R10	Local build	Built with a stick welder, drill and hand tools from common steel section	Build by a partner workshop from the published drawings	Met by design; the shell skins need plate rolls, which most fabricators have
R11	Prototype cost	At or under the USD 4,000 value-engineering	Bill of materials and receipts	Met on paper, within the target: USD 2,561 (USD 1,439 under)

ID	Requirement	Target	Verification (TRL 3 or later)	Status (SKG-CAL-001)
		target (a hypothetical control target)		

■ Requirements at risk or not met

These are stated here and decided by Amish; the options and recommendations are in SKG-DEC-001.

- **R4 (not met on paper).** Decided 1A: three people at the cranks for the hoist bring the cycle to 3.16 min, 5 % over; the timed trial settles the rest. Whether R4 is restated for three people is a new question.
- **R3 (at risk).** Decided 2B: the 3:1 tackle closes the lips with about 1.06 of the line pull; the fill in denser sand is still settled only in the test-pit bite trial.
- **R1 (at risk).** Decided 3B: the toes cut 40 mm past the ring's outer face; sinking still stops if skin friction exceeds about 1.7 kPa.
- **R9 (not met on paper by 0.1 kg).** The scraper head is 25.1 kg with the longer arms; removing the stop collar (new question) would bring it to 24.8 kg.

■ Assumptions

- Crews already use precast caisson rings of a smaller diameter below the water table; the design case is a 1.0 m ring with a 75 mm wall inside a 1.3 m lining.
- Soils at the target sites can be taken by a clamshell grab; boulders and hard rock need a separate method.
- The HatchSide tripod keeps its 150 kg rating for material handling; SinkGrab's overload limits keep the line inside it.
- Three people are available to work the capstan for a full day, two on the 240 mm two-hand handle during the hoist; a crew of four or five works the site.